Full Stack Assignment – 23

Assignment ID: FSD-23

Title: Course Enrollment Management System

Duration: 3 Hours

1️⃣ Objective

Design and develop a **Course Enrollment Management System** that allows students to enroll in courses and instructors to manage enrollments.
The system must enforce enrollment deadlines and ensure **data persistence using a database**.

2️⃣ User Roles

STUDENT

- Register and log in to the system
- View available courses
- Enroll in courses
- View enrolled courses

INSTRUCTOR

- Register and log in to the system
- Create and manage courses
- View enrolled students

3️⃣ Functional Requirements

Authentication & Authorization

- Implement user registration and login
- Use **JWT-based authentication**
- Protect all backend APIs using JWT
- Enforce **role-based access control** on APIs and frontend views

Course Management

- INSTRUCTOR can create courses with start dates
- Course data must be stored in the database
- Course details must be fetched dynamically

Course Enrollment

- STUDENT can enroll in available courses
- Enrollment data must be stored in the database

- STUDENT can view enrollment status

4️⃣ Business Rule (Critical)

⚠️ **Enrollment must not be allowed after the course start date.**

Any enrollment attempt after the start date must be rejected.

5️⃣ Technical Constraints (Mandatory)

- All course and enrollment data **must persist in a database**
- Hardcoded or in-memory storage is **not allowed**
- Backend APIs must reflect actual database state
- JWT authentication and role validation are mandatory
- Use environment variables for sensitive configuration

6️⃣ Frontend Requirements

- Login and registration pages
- Role-based dashboards for STUDENT and INSTRUCTOR
- Course listing and enrollment view for STUDENT
- Course management and enrollment view for INSTRUCTOR

UI design is not evaluated; rule enforcement and clarity matter.

7️⃣ Submission Requirements (Strict)

GitHub Repository

- Backend source code
- Frontend source code
- Proper folder structure
- Clear and meaningful commit history

Live Deployment

- Live frontend deployment link
- Live backend deployment link
- Backend must be connected to the database

README.md (Mandatory)

The README file must include:

- Project overview
- Tech stack used
- User roles and permissions

- API endpoints
- Database schema or collections
- Steps to run the project locally
- Live frontend and backend links

8 Evaluation Note

Evaluation focuses on **date-based validation, role control, and end-to-end execution**.